

Uveal Melanoma: Molecular Pattern, Clinical Features, and Radiation Response

MICHAEL C. CHAPPELL, DEVRON H. CHAR, TIA B. COLE, J. WILLIAM HARBOUR, KAVITA MISHRA, VIVIAN K. WEINBERG, AND THEODORE L. PHILLIPS

- **PURPOSE:** To characterize the clinical spectrum of class 1 and class 2 uveal melanomas and their relationship with intraocular proton radiation response.
- **DESIGN:** Masked retrospective case series of uveal melanoma patients with fine needle biopsy–based molecular profiles.
- **METHODS:** A total of 197 uveal melanoma patients from a single institution were analyzed for pathology, clinical characteristics, and response to radiation therapy.
- **RESULTS:** A total of 126 patients (64%) had class 1 tumors and 71 (36%) had class 2 tumors. Patients with class 2 tumors had more advanced age (mean: 64 years vs 57 years; $P = .001$), had thicker initial mean ultrasound measurements (7.4 mm vs 5.9 mm; $P = .0007$), and were more likely to have epithelioid or mixed cells on cytopathology (66% vs 38%; $P = .0004$). Although mean pretreatment and posttreatment ultrasound thicknesses were significantly different between class 1 and class 2 tumors, there was no difference in the mean change in thickness 24 months after radiation therapy (mean difference: class 1 = -1.64 mm, class 2 = -1.47 ; $P = .47$) or in the overall rate of thickness change (slope: $P = .64$). Class 2 tumors were more likely to metastasize and cause death than class 1 tumors (DSS: $P < .0001$).
- **CONCLUSIONS:** At the time of radiation therapy, thicker tumors, epithelioid pathology, and older patient age are significantly related to class 2 tumors, and class 2 tumors result in higher tumor-related mortality. We found no definitive clinical marker for differentiating class 1 and class 2 tumors. (Am J Ophthalmol 2012; 154:227–232. © 2012 by Elsevier Inc. All rights reserved.)

See Accompanying Editorial on page 219.

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From the California Pacific Medical Center, San Francisco, California (M.C.C.); The Tumori Foundation, San Francisco, California (D.H.C., T.B.C.); Department of Ophthalmology (D.H.C.), Department of Radiation Oncology (K.M., T.L.P.), and UCSF Helen Diller Family Comprehensive Cancer Center, Biostatistics Core, University of California, San Francisco (UCSF), San Francisco, California (V.K.W.); Department of Ophthalmology, Stanford School of Medicine, Stanford, California (D.H.C.); and Department of Ophthalmology and Visual Sciences, Washington University School of Medicine, St. Louis, Missouri (W.H.).

Michael Chappell is currently the ophthalmic plastic and reconstructive surgery fellow at The University of Washington Medical Center in Seattle, Washington.

Inquiries to Devron Char, 45 Castro St, Suite 309, San Francisco, CA 94114; e-mail: devron@tumori.org

MANY UVEAL MELANOMA PARAMETERS ARE RELATED to an increased risk of metastatic death.¹ Genomic changes, most importantly loss of a single copy of chromosome 3, are associated with an increased risk of metastasis.^{2–5} Gene expression profiling of uveal melanomas has revealed 2 distinct classes of tumors that accurately predict prognosis. Class 1 tumors are unlikely to metastasize, while class 2 tumors are more likely to metastasize and cause death.⁶ The genetic signature of class 1 and class 2 tumors can be assessed from a traditional biopsy or from a fine needle aspiration biopsy (FNAB).⁷ This gene expression profiling has been shown to have superior predictive accuracy when compared to the analysis using fluorescence in situ hybridization or array comparative genomic hybridization looking only at monosomy 3.^{8,9}

The goal of this study was to better characterize the clinical spectrum of class 1 and class 2 uveal melanomas and its possible relationship with intraocular proton radiation response.

METHODS

THIS STUDY WAS A RETROSPECTIVE CASE SERIES OF PATIENTS with uveal melanoma treated with proton beam radiation therapy between November 13, 1997 and November 12, 2010, with the last follow-up in January 2011. All patients were recruited from a single institution. Patients who did not receive proton radiation, because of primary enucleation or iridocyclectomy without radiation, were excluded from the analysis ($n = 25$). The study examined the clinical characteristics and radiation response of 197 evaluable uveal melanoma patients. Each patient underwent FNAB (185) or tumor resection with biopsy (12), followed by proton radiation. The tissues from each case underwent molecular studies and were delineated into class 1 and class 2 tumors based on the genetic characteristics at 15 loci. The specifics of this gene expression profiling have been previously discussed.⁶ In addition, cytologic or histopathologic examination was performed to determine the predominant tumor cell type. Evaluation of the tissue was performed masked to all patient, disease, and outcome features.

Clinical data and fluorescein angiographic and ultrasound characteristics were obtained from patients' initial visits. For each case, information obtained from clinical records and photographs included sex, age, ocular location,